

Doom PWAD QA Report

Run 7e9d265f-ca92-4f86-bcd3-93fdae26516e | Map E1M1 | Outcome map_completed

Executive Summary

This is an autonomous QA test of map E1M1 using an LLM agent to evaluate navigation, combat engagement, and resource balance. The test resulted in a map completion; however, the agent failed to engage any enemies due to critical resource deficiencies.

Pass/fail:

map_navigation: PASS

combat_engagement: FAIL

resource_balance: FAIL

secret_coverage: PASS

overall_verdict: PARTIAL

navigation_rationale: The agent successfully navigated to the map exit.

combat_rationale: The agent achieved 0 kills against 8 enemies, falling below the required 0.5 ratio.

resource_rationale: The health_ratio is 0 and ammo_ratio is 0, failing both minimum thresholds.

secret_rationale: There are 0 secret sectors in the map, meeting the criteria for a pass.

No technical problems encountered during this test run. Total LLM calls (5) matched total actions taken (5).

Test Item And Environment

The test evaluated E1M1 using the Freedoom1 IWAD on difficulty level 5. The map features 8 enemies (mostly hitscanners), 1 armor pickup, 0 keys, and 0 secrets within a 2528x1056 unit area. Static complexity is high due to the hitscanner-dominant enemy composition and lack of starting ammunition.

The test environment utilized the ViZDoom engine, the gemini-3.1-flash-lite LLM model, and an MCP tool interface for agent decision-making. The run was configured for a maximum of 3000 ticks at difficulty level 5.

Metric	Value	Metric	Value
Status	completed	Duration	34
Actions	5	LLM calls	5
Final HP	100	Kills	0
Position samples	214	Movement units	6348.2
Recording	/media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/7e9d265f-ca92-4f86-bcd3-93fdae26516e.mp4		

Static Analysis Context

Things	Enemies	Items	Keys	Secrets	Linedefs	Sectors
--------	---------	-------	------	---------	----------	---------

10	8	1	0	0	35	5
----	---	---	---	---	----	---

Estimated difficulty: hard. Health ratio: 0.0000. Ammo ratio: 0.0000.

Coverage Evaluation

Coverage was insufficient for a meaningful gameplay evaluation. While the agent successfully navigated to the map exit, it tested zero combat scenarios and failed to encounter any of the 8 enemies, rendering the combat and resource balance testing inconclusive.

Objectives Covered

- :
- :

Objectives Omitted

- :
- :
- :

Uncovered attributes: Map-specific scripting, sector interactions (lifts/switches), and combat mechanics were not tested as the agent exited the map immediately.

Defects And Risks

A total of 5 critical defects were detected, all related to resource balance. The most impactful defect is the total lack of starting ammunition, which prevents the player from engaging the hitscanner-heavy enemy composition.

Patterns: All defects are centered on the initial spawn and the inability to engage enemies due to zero starting ammunition, indicating a global resource balance flaw.

Severity	Priority	Type	Tick	Description
2	2	agent_observed	0	Agent-observed issue [RESOURCE BALANCE] At tick 0, position (-32.0, -224.0): Player starts with 0 bullets and no secondary weapons despite hitscanner-heavy enemy composition. Player impact: High risk of immediate death. Severity: major.
2	2	agent_observed	1	Agent-observed issue [RESOURCE] At tick 1, position (-1099, 229): Player starts with 0 bullets and no secondary weapons. Player impact: Unable to engage enemies or survive initial encounters. Severity: critical.
2	2	agent_observed	2	Agent-observed issue [RESOURCE BALANCE] At tick 2, position (31, 520): player starts with 0 bullets and no accessible weapon pickup nearby, making the initial

				encounter impossible. Player impact: cannot engage enemies, immediate death likely. Severity: critical.
2	2	agent_observed	3	Agent-observed issue [RESOURCE BALANCE] At tick 3, position (1167.93, 124.45): starting loadout contains 0 bullets for the pistol and no shell-based weapon, rendering the player unable to engage initial hitscanners. Player impact: progression blocked, immediate death likely. Severity: critical.
2	2	agent_observed	4	Agent-observed issue [RESOURCE BALANCE] At tick 4, position (90.61, -371.13): Start inventory provides zero bullet ammunition for the player's weapon. Player impact: cannot engage enemies or defend against hitscanners, preventing progression. Severity: critical.

Risk Areas

- **Starting spawn location:** high

Activity Log Summary

1) Static analysis phase, 2) Map exploration phase (ticks 0-4), 3) Combat phase (none), 4) Defect detection phase, 5) Report generation phase.

Combat phase was absent because the agent identified a resource-based softlock immediately and opted to exit the map.

Tick	Event	Position	HP	Action
4	map_exit	90.6, -371.1	100	explore

Limitations And Recommendations

The agent was unable to test combat or secondary map areas due to the immediate completion of the map.

Good Quality Areas

- **Map pathing:**